Constraint Transport and History-Dependent Successor Accessibility in IOTA: An Information-First Finite Local Physics Model

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Abstract

Biological organization can persist while the material components that realize it turn over. Using IOTA, an information-first finite local physics research model instantiated here as an artificial-life lattice simulation, we tested whether a relational constraint and an expression lineage can survive such replacement. Three strict events disconnected tracked regions after loss of an original bridge and reconnected them through a causally necessary, disjoint bridge under payload-free dynamics. Prospective ancestry showed lineage survival after death of its initial address. In 192 fresh branches, one-decision interventions reopened all 36 targeted successor routes; all six gate interventions produced an immediate child, and 28 of 30 exhaustion interventions eventually produced handoff. Vetoing 164 naturally realized unique-child events exposed an alternate continuation in 122 branches and lineage loss in 42. Sequentially vetoing the next realized child produced an observed policy cut by depth eight in 143 branches; an independent cohort reproduced cuts in 68 of 79 branches across four new source geometries. These results show, within the frozen simulator, that continuation depends on both the availability and timely realization of lawful successors and that counterfactual redundancy can recur along changed histories. To distinguish contraction from enabling, we define a signed support deformation, $\chi(R_1 : R_0) = \log[\mu(R_0)/\mu(R_1)]$. For nested uniform supports it equals forward Kullback–Leibler divergence under pruning and negative reverse Kullback–Leibler divergence under expansion. This formalizes a bounded connection to the adjacent possible, but the experiments did not enumerate complete reachable supports or establish open-ended evolution, substrate independence, or physical correspondence.

Keywords: information-first physics; artificial life; biological organization; adjacent possible; constraint transport; causal intervention; self-organization

1 Organizing Hypothesis

The organizing hypothesis of this research program is:

Information is the change in the reachable caused by interaction.

IOTA is an information-first finite local physics research model. This paper treats its stochastic lattice implementation as an artificial-life application, not as a demonstrated living system. The organizing hypothesis is to be operationalized and tested; it is not an established physical law or validated physical substrate. The narrower experimental question is: can finite local interactions produce constraints on reachable futures that persist even when the local state realizing those constraints is replaced?

The primitive experimental target is therefore the latent counterfactual structure of lawful successor events, rather than an early analogue of mass, energy, force, or a classical field. The lattice stores state; a marked relational transition graph represents which successor interactions remain possible; and coherence is provisionally the organization that preserves and transmits part of that structure across realized transitions. Familiar physical observables, if they emerge, are downstream quantities to explain rather than primitive variables to imitate. Accordingly, a proposed experiment must state which aspect of the successor structure it causally changes and measures.

1.1 Reachable supports and signed possibility-space flux

For a state s_t, define the locally reachable next-state set as

$$\mathcal{R}_x(t+1 \mid s_t) = \{(A', E') : (A', E') \text{ is admissible under } \mathcal{L}\}. \tag{1}$$

When interaction strictly reduces the reachable set, the original cardinality measure is

$$\Delta I_x = \log \frac{|\mathcal{R}_x^{\text{prior}}(t+1)|}{|\mathcal{R}_x^{\text{post}}(t+1)|}. \tag{2}$$

To distinguish pruning from enabling, let R_0 and R_1 be nonempty prior and post-interaction supports measured with the same base measure μ. Define

$$K = R_0 \cap R_1, \qquad E = R_1 \setminus R_0, \qquad L = R_0 \setminus R_1, \tag{3}$$

and the signed net-volume deformation

$$\chi(R_1 : R_0) := \log \frac{\mu(R_0)}{\mu(R_1)}. \tag{4}$$

Thus $\chi > 0$ denotes net pruning and $\chi < 0$ net enabling. For $0 < \mu(R) < \infty$, let U_R be the normalized restriction of the same measure, $U_R(A) = \mu(A \cap R)/\mu(R)$. When the supports are nested,

$$\chi(R_1 : R_0) = \begin{cases} D_{\mathrm{KL}}(U_{R_1} \| U_{R_0}), & R_1 \subseteq R_0, \\ -D_{\mathrm{KL}}(U_{R_0} \| U_{R_1}), & R_0 \subseteq R_1. \end{cases} \tag{5}$$

Counting measure gives Equation 2 in the pruning case. For nonnested supports, however, χ reports only signed net volume: it can vanish even when every outcome identity turns over.

Equation 3 must therefore accompany it; a bounded common-support diagnostic such as Jensen–Shannon divergence may be added when probability turnover is the task [20]. Empty supports require explicit set reporting or a declared absorbing outcome, not an arbitrary pseudocount.

1.2 A distribution-sensitive constraint diagnostic

Let $P_{x,\tau}^{\mathrm{prior},\pi}$ and $P_{x,\tau}^{\mathrm{post},\pi}$ be probability laws for the same declared future observable $F_{t:t+\tau}$ under policy or intervention π, before and after the focal interaction. The outcome space, measure, boundary, coarse-graining, resources, and horizon are held fixed. A candidate distribution-sensitive diagnostic is

$$\mathfrak{C}_{x,\tau}^{\pi} := D_{\mathrm{KL}}\Big(P_{x,\tau}^{\mathrm{post},\pi}\Big\|P_{x,\tau}^{\mathrm{prior},\pi}\Big)\,. \tag{6}$$

Equation 6 requires $P_{x,\tau}^{\mathrm{post},\pi} \ll P_{x,\tau}^{\mathrm{prior},\pi}$; otherwise it may diverge. It is directional, reference-relative, nonnegative, and indexed by a declared future task. It is not the signed quantity χ: $\mathfrak{C}$ detects probability reweighting, whereas χ records only net support-volume change. Neither scalar preserves the full identity and relational organization of changed futures. The complete indexed future-law family therefore remains the primary object; any aggregation over policies must be declared [19].

1.3 The adjacent frontier

Let r_ℓ map reachable microstates or trajectories to a frozen alphabet of types at scale ℓ, and let V_t^ℓ be the types already actualized by time t. For policy π and horizon τ, define the adjacent frontier

$$\mathcal{J}_{x,t,\tau}^{\pi,\ell}(s_t) := r_\ell\Big(\mathcal{R}_{x,\tau}^{\pi}(s_t)\Big) \setminus V_t^\ell. \tag{7}$$

This is not observer-free. The state, type map, policy, horizon, resources, boundary, and actualized repertoire are constitutive. With a chemical type map and a one-reaction horizon it can specialize to Kauffman's chemical adjacent possible: molecular types not yet actual but producible from the actual set [16]. No such chemical map is instantiated here. At longer horizons it is accessible novelty rather than literal one-step adjacency.

1.4 Relation to prior work

The logarithm of a finite number of alternatives is Hartley information [11]; therefore Equation 2 is a difference of Hartley support measures when its alternatives are treated uniformly, not a new information measure. Restriction-centered accounts have also identified information with limitations on possible variable values [28]. Cause–effect repertoires and effective information likewise quantify how mechanisms or interventions constrain possible states in discrete dynamical systems [1, 13].

The adjacent-possible literature is especially close prior art. Triggering urns generate correlated novelty and Heaps–Zipf regularities [25]; reinforced concept networks model discovery on a prestated

topology [15]; and dynamic scaffolding changes which possibilities become developmentally accessible [10]. Kauffman later described realized novelties as enabling constraints for typically unprestatable biological functions [17]; Patra and Kauffman sharpened that position into a proposed distinction between fixed lawful spaces and an indefinite domain of contextual novelty [22]. Numerical intervention in a Kauffman chemical model also links autocatalytic closure to complexity dynamics [8]. History-dependent accessibility, correlated novelty, constraint-enabled novelty, and causal organization-to-complexity effects are therefore not IOTA discoveries.

Geometric expansion is prior art as well. Fariña and Vallejo define a biological cosmological constant from the squared fractional expansion rate of Fisher-information volume and interpret life as a geometric transition into the adjacent possible [9]. Their sign-blind global volume rate does not provide the local $E/L/K$ ledger or a replaceable-carrier intervention, but it precludes any claim that IOTA first introduced a geometric adjacent-possible measure or generic phase transition.

Counterfactual approaches to semantic information test whether scrambling or erasure changes a system's future or viability [2, 18, 24]. Persistence of organization through component turnover is central to autopoiesis and constraint-closure accounts [3, 21, 26]. Related operational tools include local causal states for coherent structures, reachability in temporal networks, and causal ancestry in cellular automata [12, 14, 23].

At the physical-foundations level, constructor theory formulates information in terms of possible and impossible transformations [7]. A recent preprint also describes information and computation through persistent relational constraints on admissible transitions [27]. Gravity from Entropy (GfE) instead proposes a continuum gravitational action built from a geometric quantum relative entropy between a spacetime metric and a metric induced by matter and curvature [4, 5]. Its operands are metric operators, not reachable-future distributions; the possible bridge is addressed in Section 5.5.

The present novelty target is therefore narrow: complete bridge replacement, paired stochastic intervention, ordered realization cuts, and a prospective signed frontier response in this finite local computational model. Whether that combination generalizes beyond the supplied simulator remains to be tested against the native models above.

The experiments also distinguish a state-level possibility from its lifetime under ordered resolution. Let $\mathcal{R}^{(k)}$ denote the routes still admissible after event position k within a declared update. A route may be lawful in $\mathcal{R}^{(k)}$ yet disappear from $\mathcal{R}^{(k+1)}$ when a competing lawful event exhausts its source or preempts its target. This notation does not assert non-Markovian physics: the complete simulator state, random number generator state, and scheduler determine the update. It makes the causal question operational: must an ancestry-preserving successor be realized before its route closes?

2 IOTA Model and Frozen Lattice Implementation

At permanent lattice address x, write a local iota provisionally as

$$I_x(t) = (x, A_x(t), E_x(t)) \,, \tag{8}$$

where $A_x(t)$ is its integer amplitude and $E_x(t)$ collects the remaining local expression coordinates. The experiments use a 64^3 periodic lattice. The endogenous implementation state comprises the arrays `active`, A, R, V, P, `latent_A`, angular memory L, and previous-activity bookkeeping. A fixed local law $\mathcal{L}$, fixed scheduler, and pseudorandom generator define the stochastic kernel

$$S_{t+1} \sim \mathcal{L}(S_t, \xi_t; \vartheta), \tag{9}$$

where ξ_t is the declared exogenous random stream and ϑ is the frozen parameter vector. Equation 9 is a computational model definition. No identification with a fundamental physical particle or validated physical substrate is assumed.

The update is ordered rather than a purely synchronous cellular-automaton map. Tick-start active sources are visited in lexicographic array order. A source's primary vector target and scored local candidates are then considered in a deterministic candidate order. Source fields are read from the tick-start state, while target eligibility and remaining source amplitude may reflect earlier events in the same tick. Random gate draws are consumed sequentially. These scheduler semantics are part of the law and are held fixed in every paired intervention.

Each successful hard transfer debits one amplitude quantum from its source and credits the target. An inactive target with latent budget activates by combining that budget with the transferred quantum; an eligible depleted target may be recruited only through the declared residual channel. A gate with effective influence q succeeds with probability $\min(1, k_A q)$; failure changes no amplitude state. At tick end, a source whose amplitude is exhausted deactivates. The integer ledger

$$\sum_{x:\,\text{active}} A_x(t) + \sum_{x:\,\text{inactive}} \text{latent_A}_x(t) = A_{\text{total}} \tag{10}$$

is required to remain exact. Payload and vector-label updates are conditional on successful exchange and do not alter this ledger.

The principal experiments use the versioned v0.3.6 component-bridge configuration. Table 1 reports its principal values and is generated directly from the complete canonical YAML file under `output/findings/configs/`, so manuscript values cannot drift from the executable configuration.

2.1 Operational vocabulary

Iota ($I_x(t)$) A stationary lattice address together with the local attribute and expression state instantiated at that address.

Reachable set ($\mathcal{R}_x(t+1 \mid s_t)$) The set of next local states admissible from state s_t under the declared local law.

Interaction-induced information (ΔI_x) The change in the reachable caused by interaction. For strict reachable-set reduction, the current cardinality measure is the logarithm of the prior-to-posterior reachable-set size ratio.

Component	Parameter	Frozen value
System	Simulation version	`iota-rig-v0.3.6-component-bridge-residual-recruitment`
	Lattice and boundary	64^3; periodic = yes
Initialization	Reference realization	`random`; seed 42; 16 active iotas
	Initial expression ranges	$A \in [1,7]$; $R = 7$; $P \in [0,7]$
Local law	Neighborhood and vectors	neighbor mode `cube26`; vector mode `cube26`; 64 vectors
	Vector behavior bands	move 0–25; reflect 26–51; diffuse 52–63
	Transfer and conservation	$k_A = 5.0$; range coupling = 1.0; bond probability = 1.0; valence 8; conservation `hard`
	Decay and deactivation	decay `inverse_square`; deactivation `exhaustion`; vacuum amplitude 2
	Payload dynamics	takeover = 0.5; same/cross couplings = 1.0/1.0
	Residual recruitment	probability = 0.02; secondary-only = yes
Angular control	Received rotation	probability = 1.0; maximum offset 3; preserve behavior = yes
	Angular memory	enabled = yes; decay/coupling/retention = 0.9/0.5/0.5; cap = 2.0
	Residual angular bias	enabled = yes; strength = 2.0; gate `depleted_interface`; mean-preserving = yes
	Residual-bias score	`source_target_L_contrast_component_bridge`
	Nonzero geometry weights	component bridge cube26 0.25; local bridge face6 0.05; flat front -0.05; edge extension -0.05; remaining 5 classes 0
Run	Registered horizon	2048 ticks; snapshots every 256; metrics every 256

Table 1: Principal frozen parameters of the canonical v0.3.6 component-bridge reference configuration. Values are generated directly from the complete registered YAML source.

Adjacent frontier ($\mathcal{J}^{\pi,\ell}_{x,t,\tau}$) For a declared state, policy, horizon, resource boundary, scale-specific type map, and previously actualized type repertoire, the types represented among reachable futures that have not yet been actualized. It is a bounded and observer-indexed accessibility construct, not an observer-free or open-ended phase space.

Possibility-flux ledger ((K, E, L)) For prior and post-interaction reachable supports defined on the same outcome space, the ordered decomposition into retained outcomes K, enabled outcomes E, and lost outcomes L. The ledger preserves outcome identity when a scalar net-volume change would hide turnover.

Signed support deformation ($\chi(R_1 : R_0)$) The logarithm of prior reachable-support measure divided by post-interaction reachable-support measure, using one declared base measure on finite nonempty supports. Positive values denote net pruning, negative values denote net enabling, and zero does not exclude complete outcome turnover.

Coherence A connected active structure measured using the explicitly declared neighborhood connectivity. It is not assumed to be a fundamental physical object.

Bridge (B) A set of active lattice cells whose removal changes reachability between tracked regions in the audited state.

Strict constraint transport Loss and restoration of reachability between the same tracked predecessor regions through a causally necessary successor bridge disjoint from the predecessor

bridge.

Payload-free ($P = 0$) World.P is zero at every observed offset in the audited continuation.

Ledger exactness The experiment's conservation ledger has zero recorded discrepancy at every audited offset.

Predecessor address lineage A conservative lineage that remains alive while at least one original stationary iota address of the predecessor region remains active; later reactivation after complete address loss does not restore the terminated lineage.

Prospective expression lineage A non-resurrecting lineage propagated only when a successful transfer from a current lineage address newly activates its target in that tick; retained active addresses remain members, already-active targets do not inherit identity, and ambiguous dual claims invalidate a strict result.

Expression-handoff susceptibility (p_{handoff}) Among continuation branches from a fixed state in which the original singleton address becomes inactive within the declared horizon, the fraction retaining a nonempty, collision-free prospective singleton expression lineage immediately after that first deactivation.

Candidate lineage-continuation rule A frozen mapping from features available immediately before original-address deactivation to the presence or absence of a nonempty prospective expression lineage immediately afterward; a predictive marker is not treated as a primitive law unless it is both sufficiently sensitive and independently replicated.

Temporal continuation gate An ordered event decision whose resolution before source termination changes whether an ancestry-preserving successor is realized; it is identified by a frozen intervention that changes that decision while leaving the pre-intervention state and consumed random draw fixed.

Opportunity lifetime (τ_{opp}) The ordered event or tick interval during which a specified successor transition remains admissible, ending when that transition is realized or a declared lawful event makes the route unavailable.

Realization redundancy For a factual ancestry-producing event selected before intervention, temporal redundancy occurs when vetoing that event leads to another ancestry-preserving child within the declared follow-up. The selected event is branch-necessary when the address instead dies without inherited continuation; continued source activity at the horizon is censored.

Sequential realization cut depth (D_{seq}) Under a frozen retained-amplitude veto policy and fixed follow-up, the smallest depth d along a branch's prospectively committed selector chain at which vetoing the first d ancestry-producing events causes the original address to deactivate

without a live collision-free descendant. Handoff after the administrative maximum is right-censored above that depth; continued original-address activity at the horizon is source-survival censored. The depth is prefix-minimal for this ordered policy, not a graph-theoretic minimum over arbitrary intervention sets.

Latent counterfactual structure The marked relational structure of lawful successor events still realizable from a declared complete state under the declared law, including their conditional order, compatibility, and closure. It is not the single factual trajectory, a scalar field, or a claim that every successor is simultaneously available.

Frozen rooted graph-role signature For Finding 075, the tuple of face6 and face18 active-neighbor counts, face6 and cube26 neighbor-component counts, and the sorted induced cube26-degree sequence among a target's active cube26 neighbors, conditioned on exact cube26 shell-one and Chebyshev shell-two occupancy. Equality of this signature is a declared matching representation, not graph isomorphism or an established sufficient statistic for successor dynamics.

Coherence-conditioned transition geometry The organized, potentially anisotropic and history-dependent structure of lawful next transitions conditioned on a focal coherence, its marked relational neighborhood, and its recent causal history. The term does not presuppose a scalar, vector, tensor, Euclidean field, force, gravity analogue, or independently confirmed predictive representation.

Successor openness (O_{succ}) For a declared pre-step state and radius-two basin, the sum of frozen transfer probabilities over tick-start-eligible inactive-target activation edges in the marked hard-transfer candidate graph, accompanied by the number of distinct activation targets. It is not an entropy, an exhaustive multi-tick reachable set, or an independently confirmed predictor.

Repair affordance (A_{repair}) For a declared pre-step state and radius-two basin, the sum of frozen transfer probabilities over eligible depleted-target secondary edges classified as component_bridge_cube26 or local_bridge_face6, accompanied by distinct target and source counts. It measures one-step bridge/reclosure accessibility under the declared representation, not realized repair or a universal counterfactual quantity.

Factorial interaction ($I_{T\times M}$) For natural, topology-only, marks-only, and joint states, the declared instrument contrast joint minus topology-only minus marks-only plus natural. A nonzero value establishes non-additivity of the instrument under those interventions; by itself it does not establish a downstream causal or predictive effect.

3 Constraint-Transport Test

Finding 060 replayed the 36 Class B bridge-relocation events identified by the preceding class-matrix experiment. The audit required a role-bearing predecessor bridge and tracked the exact predecessor regions through bridge loss and reconnection. Pairwise reachability and shortest-path accessibility were measured before loss, during the split, and after rebridge. The successor bridge was then removed to test whether it was causally necessary.

Let $C_{UV}(t)$ equal one when tracked regions U_t and V_t are mutually reachable and zero otherwise. A strict event requires

$$C_{UV}(t_{\mathrm{pre}}) = 1, \qquad C_{UV}(t_{\mathrm{split}}) = 0, \qquad C_{UV}(t_{\mathrm{post}}) = 1, \tag{11}$$

together with disjoint predecessor and successor bridges, $B_{\mathrm{pre}} \cap B_{\mathrm{post}} = \varnothing$, and loss of restored reachability after removing B_{post}.

The analysis code and exact source tables are linked from Section 7.

3.1 Expression ancestry and terminal-event intervention

A prospective expression lineage retains an active address and inherits a newly activated target only from a successful transfer whose source belonged to that lineage before the update. Already-active targets do not inherit identity, terminated lineages do not resurrect, and competing claims to one child mark a collision.

For Finding 067, a default-off observer recorded selected transfer candidates in evaluation order, including target state, transfer probability, random draw, and effective decision. Paired observer-free and sham-intervention worlds were required to remain exactly equal in every state array and RNG state. Each causal replay consumed the natural random draw but changed exactly one decision: permit a failed inactive-successor gate, veto the non-child transfer spending the source's final amplitude, or veto the earlier activation preempting a target. The primary mediators were route reopening and immediate inherited-child creation; eventual handoff at first address death was evaluated separately.

3.2 Independent-geometry holdout

The independent screen exhausted a frozen 16×8 matrix: unused initialization seeds 57–72 and eight distinct continuation streams per seed. Every Class B candidate received the unchanged strict transport audit. Before terminal outcomes were examined, the source manifest retained at most the lexicographically first eligible singleton transport per initialization seed. Each selected source was then branched 32 times at rebridge, rather than at a later terminal-aligned state, and evaluated with the same intervention policy. Initialization seed is the independent geometry unit; branch outcomes are conditional stochastic replicates.

3.3 Reverse necessity and redundancy test

Finding 068 separated factual selection from intervention. First, all 256 fresh factual branches were completed and every collision-free, single-address, natural child event was written to a selector manifest. The manifest stored the candidate and gate ordinals, source, target, amplitude, target state, probability, and random draw and was committed before replay. Every selected factual event had exactly one inherited child.

The replay stage accepted only the committed manifest hash. It reproduced an exact sham, matched every frozen event field, consumed the natural draw, and forced that one successful child gate to fail. Each branch was followed for 128 offsets after intervention or until first original-address death. Alternate child timing (same tick, delayed, or none) and address endpoint (handoff, lineage loss, or source-survival censoring) were recorded as independent axes.

3.4 Sequential retained-amplitude cut protocol

Finding 069 extended the committed Finding 068 cohort with a staged sequence of at most eight vetoes. At depth d, the complete selector prefix through $d-1$ was replayed from the common branch state. For every branch that still handed off, the first next ancestry-preserving child was joined to its successful candidate-ledger row. Its tick and gate ordinal, source, target, candidate step and order, available amplitude, target state, probability, and random draw were written to a complete manifest and hashed and committed before the depth-d gate was vetoed. Every later replay had to reproduce all earlier selector identities and natural gate passes exactly. A veto consumed the frozen random draw but suppressed the transfer, thereby retaining at the source the amplitude that the natural realization would have spent.

Let $H_b^{(d)} = 1$ when branch b retains a collision-free descendant at first original-address death after the first d committed vetoes, and let $H_b^{(d)} = 0$ when that address dies without one. The measured depth is

$$D_b = \min\{d \in \{1, \ldots, 8\} : H_b^{(d)} = 0\}. \tag{12}$$

This is a prefix-minimal depth under the frozen first-next-child and retained-amplitude policy. It is not a minimum over arbitrary event subsets or state interventions. Every stage used the same endpoint window: 128 offsets after the depth-1 intervention. A depth-8 handoff was right-censored above the protocol maximum; an original address still active at the fixed horizon was separately source-survival censored. Collision, payload, or conservation failures were protocol failures, not post-treatment exclusions.

3.5 Independent ordered-cut replication

Finding 070 froze a 256-case screen from 32 previously unused initialization seeds before continuation outcomes. The lexicographically first strict eligible source per seed yielded four source geometries. Thirty-two factual branches per source were completed, and every collision-free natural continuation

with exactly one inherited child was committed before any veto. The identical Finding 069 selector rule, retained-amplitude intervention, 128-offset endpoint, and depth-eight cap were then applied. Replication was preregistered as at least five eligible selectors per source and a depth-1 handoff in at least two newly selected sources; exact depth-1 matched shams and the full selector-hash chain remained required protocol checks.

4 Results

In three controlled replay events across two seeds, payload-free Iota dynamics restored the same tracked connectivity relation through a causally necessary successor bridge having no cells in common with the predecessor bridge.

Of the 36 Class B events replayed, 12 had a role-bearing predecessor bridge and 3 passed the strict criterion. The positives occurred across 2 seeds. For all three positives, bridge-path Jaccard overlap was zero, reachability changed from one to zero to one, successor removal returned reachability to zero, and pre- and post-event shortest-path length was two. Payload was zero and the conservation ledger was exact.

Table 2 lists the three strict events and their disjoint predecessor and successor bridges.

Seed	Sample	Start	Rebridge	Predecessor	Successor	J
46	7	61	65	17:27:24	16:28:25	0.0
46	7	420	424	39:16:19	40:17:20	0.0
56	3	106	112	14:4:5	15:5:6	0.0

Table 2: Strict constraint-transport events in Finding 060. J is bridge-path Jaccard overlap.

4.1 Post-transport address persistence

Finding 061 followed the original predecessor addresses after rebridge. The three strict transports preserved their restored relation for 9, 3, 51 recorded offsets, whereas all 9 role-bearing failures had zero relation lifetime at nominal rebridge. Each strict-positive address lineage terminated when its singleton predecessor address became inactive. No second continuous strict transport was observed through offset 1024.

4.2 Prospective expression lineage

Finding 062 instrumented successful hard-transfer activations and verified that the observer left paired simulation and RNG states exactly unchanged across 94 observed transfers. Under the preregistered source-to-new-target ancestry rule, the three strict-event relation lifetimes were 170, 601, 51 offsets. 2 of the three singleton expressions continued after their original address became inactive; the longer remained alive at the offset-1024 censoring horizon. All nine role-bearing failures remained at zero lifetime, no lineage collisions occurred, and no second strict transport was observed.

4.3 State-conditioned handoff susceptibility

Finding 063 branched 32 continuation RNG streams from each of the three strict-rebridge states. Handoff rates were 50.0%, 90.6%, 87.5%, respectively. Two-sided Fisher exact comparisons gave $p = 0.000752$ for the first versus second seed-46 state, $p = 0.00251$ for the first seed-46 versus seed-56 state, and $p = 1.0$ for the two high-susceptibility states. All 96 original addresses deactivated within the declared horizon, no lineage collision occurred, payload remained zero, and the ledger remained exact.

4.4 Held-out compact-rule test

Finding 064 trained a preregistered decision stump on 48 terminal-aligned branches and evaluated it once on 48 held-out branches. The selected rule was singleton-lineage size greater than 1.5 immediately before original-address death. Held-out sensitivity was 0.361, specificity was 1.000, and balanced accuracy was 0.681 ($p = 0.0210$, two-sided Fisher exact). The marker had no false positives but missed 23 of 36 handoffs; it is therefore treated as a sufficient state marker, not a general continuation law.

4.5 Fresh confirmation of a static-geometry rule

Finding 065 excluded endpoints with preexisting descendants, selected a new stump using only the original discovery rows, and evaluated it on 96 fresh continuations. The frozen rule predicted child creation for at most one active cube-26 neighbor. Among 71 eligible single-address endpoints, held-out balanced accuracy was 0.582 and the association was not significant ($p = 0.240$, two-sided Fisher exact). The preregistered static terminal features therefore did not yield a confirmed general continuation rule.

4.6 Ordered terminal events and a causal continuation gate

Finding 066 instrumented every selected hard-transfer candidate on the terminal tick for 96 fresh branches. The observer passed exact state-and-RNG equivalence and recorded 20,748 terminal candidate events. The ordered ledger classified all 96 outcomes without collisions: 57 death-tick child activations, 16 preexisting-descendant continuations, 19 source-exhaustion deaths, three transfer-gate deaths, and one target-preemption death. In a preregistered replay of one gate-failure branch, permitting exactly one failed successor gate while still consuming its original random draw changed the outcome from lineage death to one collision-free ancestry-preserving child. This supports a bounded causal claim: continuation can depend on realizing an eligible successor before the source expends its terminating amplitude.

4.7 Prospective causal replication and independent geometries

Finding 067 froze three one-decision policies before evaluating 192 fresh branches from the original source states. Permitting the first failed gate to an inactive successor produced an immediate

inherited child in 6/6 cases. Vetoing the successful non-child transfer that would spend the final source amplitude reopened the route in 30/30 cases; three formed a child immediately and 28 eventually handed off. Exact matched shams, one-event matching, payload absence, conservation, and collision freedom held throughout.

A complete screen of 128 continuations from 16 unused initialization seeds found four singleton-eligible strict transports across two seeds and froze one primary source per seed before terminal outcomes. In 64 branches from these 2 independent geometries, seed 71 produced eight successor-gate deaths; all eight interventions produced immediate children and handoff. One seed-58 exhaustion intervention reopened the route but did not eventually hand off, while one seed-71 preemption intervention restored target availability and was followed by handoff four offsets later. These results separate two causal conditions: route availability and successor realization. Reopening opportunity is causal, but it is not by itself sufficient for continuation.

4.8 Necessity and temporal redundancy of realization

Finding 068 completed 256 fresh factual branches and committed 164 unique-child selectors before any veto outcome. Suppressing the selected child produced an alternate child on the same tick in 5 branches, after a delay in 117, and never in 42. All 122 branches with an alternate child handed off at first address death; all 42 branches without one lost the lineage. No source-survival endpoint was censored.

The selected target was the first alternate target in only 17 cases; 105 alternatives used a different address. Source capacity was retained immediately after veto in 161/164 branches. Both temporal redundancy and branch-level necessity occurred in every represented initialization seed. All selected events were suppressed, all shams and selector identities were exact through the original draw, and no collision, payload, or conservation failure occurred.

4.9 Sequential realization cut depth

Finding 069 began with the same 164 factual unique-child branches and froze each next-depth selector manifest before applying the newly selected veto. The observed cut counts at depths 1 through 8 were 42, 20, 22, 17, 11, 13, 13, and 5, respectively. Thus 143 branches reached first original-address death without a descendant under an observed policy prefix. After all eight vetoes, 19 branches still handed off and are right-censored above the protocol maximum; 2 additional branches retained the original source through the 128-offset window and are source-survival censored rather than assigned a cut depth.

The complete depth-specific risk sets and endpoints are reported in Table 3.

Across the adaptive chains, 648 distinct branch-selector identities were committed. Replaying every complete prefix produced 2142 gate-override executions. 80 branches selected at least one target address that had appeared earlier in the same selector chain, showing that route replacement may revisit an address as well as move to a new one. All selector hashes, parent-chain hashes, branch nesting, inheritance joins, and event identities verified. There were 0 lineage collisions; payload

Depth	Risk set	New cuts	Handoffs	Source-censored
1	164	42	122	0
2	122	20	102	0
3	102	22	80	0
4	80	17	63	0
5	63	11	52	0
6	52	13	39	0
7	39	13	26	0
8	26	5	19	2

Table 3: Finding 069 ordered realization cut outcomes. Each depth's risk set is the preceding depth's confirmed handoffs.

remained zero and the conservation ledger remained exact in every stage. These counts estimate the ordered, retained-amplitude policy response, not a globally minimal successor cut.

4.10 Independent replication across initialization geometries

Finding 070 selected 4 new source geometries and froze 79 factual unique-child continuations. The observed depth-1 through depth-8 cut counts were 23, 15, 7, 9, 4, 3, 5, and 2, respectively. 68 branches had an observed cut, 10 still handed off after eight vetoes, and 1 was source-survival censored at depth seven.

Table 4 reports the corresponding independent risk sets and endpoints.

Depth	Risk set	New cuts	Handoffs	Source-censored
1	79	23	56	0
2	56	15	41	0
3	41	7	34	0
4	34	9	25	0
5	25	4	21	0
6	21	3	18	0
7	18	5	12	1
8	12	2	10	0

Table 4: Finding 070 independent ordered-cut replication. Each depth's risk set is the preceding depth's confirmed handoffs.

Every source geometry produced at least one depth-1 handoff, exceeding the preregistered two-source criterion. The chains contained 286 committed selector events and 923 override executions; 38 branches reused a target. One observed cut contained a transient inherited child that disappeared before address death. All hash, join, sham, payload, conservation, and collision checks passed.

Later registered Findings 074–079 concern a separate transition-geometry and representation-development program. They remain archived in the repository but are outside the claim set analyzed in this article.

5 Discussion

5.1 Constraint transport is a role claim, not material identity

The strict positives show that a connectivity relation can, in a restricted event class, outlast complete replacement of the cells that instantiate its bridge. The particular bridge is therefore not sufficient to identify what persists. The invariant candidate is the restored reachability relation between tracked region descendants. This is consistent with the organizing hypothesis: interaction can alter reachable collective states, and a constraint on those states can be re-instantiated in a different local realization. Three events do not, however, establish a general reachable-set law or substrate-independent organization.

Prospective expression ancestry adds an event-resolved mechanism. A lineage can survive death of its stationary address when an ancestry-preserving successor is realized while the route remains available. A route may close because a gate fails, a source spends its final amplitude elsewhere, or another source activates the target first. In this model, *before* is therefore causal and ordered rather than merely chronological: changing one identified decision changes whether the route remains open or a child is realized.

The intervention results separate availability from realization. Every exhaustion intervention in the prospective same-state cohort reopened the route, yet two did not later hand off; the independent seed-58 case likewise reopened opportunity without rescue. Persistence requires both an available lawful route and timely realization. No payload transport is needed for the operational lineage used here.

5.2 Counterfactual redundancy is history-dependent

The naturally realized child was not generally unique in the reachable structure. Although every selected factual branch expressed exactly one child, veto exposed an alternate in 122 of 164 counterfactuals, usually at a different target and often after a delay. The selected realization was temporally redundant in those branches. In the remaining 42, no alternate appeared and the lineage died at address loss, establishing branch-level but-for necessity under the declared intervention.

Sequential vetoes show that redundancy can recur along several altered histories. Cuts occurred at every tested depth from one through eight; 143 of 164 branches lost the lineage after some committed prefix. The measured depth is robustness to the frozen first-next-child, retained-amplitude policy, not a static stock of interchangeable children or a globally minimal graph cut. Each veto preserves amplitude and changes the subsequent lawful path, so a later successor can become available only in the counterfactual history.

The independent cohort reproduced the policy response across four prospectively screened source geometries: all four retained continuation after the first veto, and 68 of 79 branches cut by depth eight. This supports geometric recurrence under the frozen law and scheduler. It does not establish scheduler invariance or an exhaustive topology.

5.3 Registered cross-domain extension: a bounded Dirac role

The same prospective discipline was also applied in a separate quantum-Dirac construction study [6]. An ordered product walk and an established direct symmetry-derived quantum cellular automaton were frozen in separate registrations and tested against a common, bounded $1+1$-dimensional Dirac operational-role core. Both recorded positive development and single-use confirmation classifications for their intact low-momentum witnesses, causal and symmetry checks, and pre-outcome analytic leakage bounds.

A hostile post-result audit then found a shared factor-of-two error in the direct-QCA empirical π-sector predicate: the predicate selected an empty sector in both implementations. The affected measured-zero claims were withdrawn, while the correctly defined analytic bound and other registered evidence survived. An exact audit also showed that the one-excitation laws are related by a half-cell time-frame conjugacy after mass-angle remapping, so the study does not establish independent architecture classes or a resource advantage. Its defensible result is qualified constructive sufficiency for the shared bounded role core. Here it is a methodological application of the same freeze–test–audit discipline, not evidence for biological organization, quantum emergence from IOTA, the adjacent frontier, or χ; the detailed result and limitations remain in the dedicated manuscript.

5.4 Relation to the adjacent possible

These veto experiments resemble adaptive exposure of the adjacent possible, but the current evidence does not measure an adjacent frontier. Alternate child *addresses* are not automatically novel types. No type map r_ℓ, actualized repertoire V_t^ℓ, or complete support $\mathcal{R}^\pi_{x,\tau}$ was frozen before the existing outcomes. The 122 alternate continuations therefore establish policy- and history-dependent successor accessibility within a finite prestated simulator, not new state-space dimensions, open-ended evolution, or ontic creation of possibilities.

The prospective extension is to freeze those objects and apply the same carrier formation, ablation, replacement, restoration, and veto interventions. A useful test would report the enabled, lost, and retained identities in Equation 3 and predict, before held-out runs, a cross-scale change

$$\chi_{\mathrm{micro},1} > 0, \qquad \chi_{\mathrm{role},h} < 0 \quad (h \geq h_\star), \tag{13}$$

where $h_\star$ is estimated from independent gate-lifetime or cut-depth data. The immediate intervention may prune microstate routes while a successfully transported role enables more long-horizon continuations. This crossover was not estimated in the reported experiments; it is a preregisterable prediction.

5.5 Candidate bridge to Gravity from Entropy

The signed identity in Equation 5 supplies a precise algebraic bridge target, not a derivation of gravity. In Gravity from Entropy (GfE), the local Lagrangian is

$$\mathcal{L}_{\mathrm{GfE}}(x) = -\operatorname{Tr}_F \ln \boldsymbol{\Theta}_x, \qquad \boldsymbol{\Theta}_x = \widetilde{\mathbf{G}}_x \widetilde{g}_x^{-1}, \tag{14}$$

where $\boldsymbol{\Theta}$ is positive and the operands are higher-order metric operators rather than probability distributions [4, 5]. In a finite flattened representation this is the log-determinant contrast $-\ln\det_F \boldsymbol{\Theta}_x$, where $\det_F$ denotes the ordinary determinant on the declared finite flattened common space.

A bridge theorem would require positive future-law operators $\mathbf{Q}_x^{\mathrm{prior}}$ and $\mathbf{Q}_x^{\mathrm{post}}$, constructed independently from the IOTA transition law, such that

$$\begin{aligned} \boldsymbol{\Theta}_x &\overset{?}{=} (\mathbf{Q}_x^{\mathrm{prior}})^{-1/2} \mathbf{Q}_x^{\mathrm{post}} (\mathbf{Q}_x^{\mathrm{prior}})^{-1/2}, \\ \det_F \mathbf{Q}_x &= a\, \mu(\mathcal{R}_x), \qquad a > 0 \text{ common to both conditions.} \end{aligned} \tag{15}$$

Under those algebraic assumptions, the requested bridge formula is

$$\boxed{\mathcal{L}_{\mathrm{GfE}}(x) = \log \frac{\mu(\mathcal{R}_x^{\mathrm{prior}})}{\mu(\mathcal{R}_x^{\mathrm{post}})} = \chi(\mathcal{R}_x^{\mathrm{post}} : \mathcal{R}_x^{\mathrm{prior}})}. \tag{16}$$

For finite nested supports under counting measure, this equals ΔI_x and forward KL divergence under pruning, but equals negative reverse KL divergence under expansion. The signed scalar aligns the sign convention with the GfE mode parameterization; it does nothing more by itself. A conditional mode dictionary would read

$$\chi_{x,k} \widehat{=} c_k := \frac{\delta s_k}{\delta v}, \qquad \mathcal{G}_k = e^{\chi_{x,k}}, \qquad \frac{\delta \epsilon_k}{\delta v} = e^{\chi_{x,k}} - 1 - \chi_{x,k}. \tag{17}$$

The question mark remains physical: common operator dimension, positivity, units, normalization, metric meaning, covariance, functional variation, and a continuum limit have not been derived. Nonnested supports can mix pruning and enabling even when $\chi = 0$, so $E/L/K$ identities remain necessary.

The bridge must also be distinguished from the published biological cosmological constant Λ_B, which uses the squared fractional expansion rate of Fisher-information volume [9]. That construction is direct prior art for geometric expansion and a phase-transition interpretation; Equation 16 instead proposes a local signed carrier-intervention contrast. Neither construction confirms the other.

The current simulator is therefore a causal testbed only for constructing and challenging $\mathbf{Q}_x$. A credible continuation must freeze the operator and measure map, predict the sign crossover and $h_\star$ on held-out systems, and outperform complete IOTA dynamics, native adjacent-possible models, Λ_B, native GfE, and ordinary branching or percolation explanations.

6 Limitations and Falsification Frontier

The structural foundation remains three strict events across two original initialization seeds, all preserving a large-region/singleton relation. The independent screen adds two primary geometries

for the one-decision mediator tests, but only one contains a replicated gate-failure cohort of at least five cases. The independent exhaustion and preemption cohorts contain one case each and support only case-level mediator claims.

The ledger covers candidates selected by the declared hard-transfer routine. Candidate selection and lexicographic source scheduling are not yet separately intervened upon and may be implementation-specific. Forcing a failed gate to pass is an efficacy test, not an estimate of its natural probability. The original draw is consumed and paired worlds are identical through intervention, but later random-number consumption can diverge after their states diverge.

The reverse-veto cohort conditions on factual natural continuation and therefore estimates robustness, not the prevalence of necessity across all branches. Vetoing a successful child retains source amplitude and intentionally exposes alternate routes. Inherited realization is part of the operational lineage definition, so the exact alternate-child/handoff correspondence is not an independent predictive association. Five source states represent four initialization seeds.

Sequential depth is prefix-minimal only for the frozen first-next-child policy. Because each suppressed transfer leaves amplitude at its source, successive counterfactual worlds do not expose a fixed list of factual routes. The result is neither a graph-theoretic minimum cut nor evidence that each earlier prefix member is individually necessary. The protocol also truncates two axes: right-censoring above depth eight and source-survival censoring at the common follow-up endpoint. Independent replication remains conditional on factual unique-child continuation and supports recurrence, not a scheduler-invariant topology.

The adjacent-possible interpretation is prospective. No outcome-type map, actualized repertoire, complete reachable support, $E/L/K$ ledger, χ, or adjacent frontier was estimated in the reported cohorts. Alternate addresses need not be alternate types. The selected branches are censored and policy-dependent, sequential vetoes retain amplitude and alter history, and the finite law and alphabet cannot demonstrate open-ended phase-space creation. Even when measured, χ alone can miss complete identity turnover in nonnested supports.

Indefinite persistence, iterated strict transport, a necessary-and-sufficient continuation law, uniqueness to this update law, and physical correspondence remain unestablished. Further falsification should extend the frozen horizon and depth, randomize or compare selector policies and source schedules, fit complete state and native adjacent-possible baselines, and enumerate small intervention subsets on held-out branches. Mass, energy, force, matter, and physical-field correspondence remain downstream hypotheses rather than demonstrated substrate primitives.

The relative-entropy relations in Equations 6 and 5 are mathematical diagnostics, not empirical validation of the organizing hypothesis. The GfE map adds further untested requirements: positive operators on a common space, a derived metric and volume interpretation, units, normalization, sector control, covariance, functional variation, a continuum limit, and robustness across laws, schedulers, topologies, boundaries, and representations. No GfE scaling, gravitational transition, or observational signature has been derived or tested by this simulator. A future geometric claim must also distinguish the signed carrier response from the existing Fisher-volume Λ_B construction.

7 Reproducibility and Evidence

The paper uses generated rather than manually copied numbers. Canonical claims and definitions are in `evidence/claims.yaml` and `evidence/definitions.yaml`. Each `evidence/manifests/GC-NNN.json` records the exact path, byte count, and SHA-256 digest of supporting evidence and code; raw artifacts remain under `output/findings/`.

Finding 069's cases, prospective selector manifests, and aggregate tables are in `output/findings/finding069/`; its claim manifest identifies both implementations. Before reporting cut assignments, the synthesis verifies selector and parent-chain hashes, contiguous depths, nested risk sets, event identities, and safety invariants.

Finding 070's independent screen, factual freeze, eight staged risk sets, and replication synthesis are in `output/findings/finding070/`. Its synthesis revalidates the fixed source and selector hashes, nested risk sets, exact selector-to-case joins, shams, endpoints, and safety invariants.

Later Findings 074–079 remain registered in the same evidence system but are not part of this article's claim set.

`make verify` rebuilds both eight-stage syntheses, regenerates derived files, and validates expectations, provenance, paths, and freshness. `make paper` then compiles the review PDF and produces a stable PDF, SHA-256 checksums, build manifest, and self-contained LaTeX source archive under `output/pdf/` and `output/submission/`.

8 Conclusion

In this artificial-life application of IOTA's finite local model, a relational connectivity role survived complete replacement of its original bridge, and an expression lineage survived loss of its initial address through lawful successor realization. Paired gate, exhaustion, preemption, and reverse-veto interventions show that continuation depends on both route availability and timely realization. Repeated vetoes and an independent four-geometry cohort further show that counterfactual redundancy can recur along changed histories rather than residing in one static reserve.

The result is deliberately bounded. It demonstrates causal constraint transport and history-dependent successor accessibility inside a frozen stochastic simulator. It does not demonstrate an unprestatable adjacent possible, open-ended evolution, substrate independence, or emergent gravity. The signed support deformation χ and its pruning–enabling KL duality provide a prospective way to test those future-space changes without conflating contraction and expansion. The separately registered Dirac-QCA construction shows that the same prospective method can survive a defect audit with a narrowed cross-domain claim; it does not extend the present biological result. The decisive next experiment is therefore not another retrospective analogy: it is a preregistered carrier intervention that predicts enabled, lost, and retained future identities and a held-out micro-pruning to role-enabling crossover beyond complete native models.

Data and code availability

The simulator, frozen configurations, intervention code, evidence manifests, generated tables, and manuscript sources are publicly available at https://github.com/fmertz55/iota. Each registered result links its source artifacts by path, byte count, and SHA-256 digest.

Ethics statement

This theoretical and computational study used no human participants, human tissue, animals, or field samples; ethics approval and consent were not applicable.

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Douglas Clark: Conceptualization; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Writing – original draft; Writing – review and editing; Project administration.

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During preparation of this work, the author used OpenAI Codex to assist with literature discovery, code and manuscript review, and language editing. The author verified the cited sources, analyses, code, and text, revised the material as needed, and accepts full responsibility for the content.

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